Peter Refaat

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EDUCATION

Ain Shams University, Faculty of Computer and Information Sciences — Cairo, EG
Bachelor's in computer science — Oct 2027

- Relevant Coursework: Object Oriented Programming, Data Structures, Analysis and Design of Algorithms, Operating Systems, Data Bases, Artificial Intelligence, Computer Networks, Compiler Design, Software Analysis and Design.
- GPA: **3.89/4.00**

King Fahd Model Language School, Upper Secondary Education — Cairo, EG
National Upper Secondary Education Certificate (Mathematics Section) — Sep 2023

- Score: **399/410** (**97.3%**)

EXPERIENCE

acmASCIS — Cairo, Egypt
Competitive Programming Instructor / Head of Resources Team — Oct 2024 – Present

- Mentoring students of all levels in my college through both offline and online sessions.
- Teaching and explaining programming concepts ranging from introductory programming to advanced topics such as Number Theory, Graphs, Dynamic Programming, and Data Structures.
- Guiding students in problem-solving techniques and preparing them for competitive programming contests.

SKILLS

- **Programming Languages:** C++, Java, Python & JavaScript.
- **Soft Skills:** Problem Solving, Mentoring & Teaching.
- **Competitive Programming:** Solved **3500+** problems across platforms such as Codeforces, LeetCode, AtCoder
- , and others; achieved Expert rating on Codeforces and competed in ECPC and ACPC regional contests.

HONORS AND AWARDS

ECPC
Silver Medalist — Aug - 2025

- Ranked as one of the **top 10** teams in Egypt over **264** participating teams and qualified as one of the Egyptian teams representing Egypt in the **2025 ICPC Africa & Arab Collegiate Programming Championship (ACPC)**.

ACPC
Finalist — Dec – 2025

- Competed among **148** regional teams in **2025 ICPC Africa & Arab Collegiate Programming Championship (ACPC)** representing Egypt at the regional level.

PROJECTS

FCIS Operating System (FOS) https://github.com/Peter-Refaat/FOS-OS — Dec - 2025

- Implemented **dynamic memory allocation** with power-of-two block segmentation and free list management for efficient small allocations.
- Built **kernel heap management** with separate block/page allocators and custom-fit strategy.
- Developed **page fault handling** with disk-to-memory page loading, working set tracking, and multiple replacement algorithms.
- Implemented **synchronization primitives** including sleep locks and semaphores with proper blocking/wakeup systems.
- Built **priority-based CPU scheduler** with round-robin execution and starvation prevention mechanisms.
- Created **user-space memory management** supporting malloc/free operations with delayed allocation via page faults.

Image Segmentation https://github.com/Peter-Refaat/Image-Segmentation — May - 2025

- Implemented **graph-based segmentation** using **disjoint set union (union-find)** and **Kruskal's MST** for region merging with 8-connected neighborhoods.
- Optimized edge processing with **counting sort algorithm** for **O(n)** edge weight sorting and **parallelized** R/G/B channel segmentation.
- Built **multi-channel processing** system with **parallel channel segmentation** and intersection logic for color image handling.

Mini-Wasalny https://github.com/Peter-Refaat/Mini-Wasalny — May - 2025

- Built Qt desktop application for **visualizing graph algorithms (DFS, BFS, Dijkstra, Floyd-Warshall)**
- Implemented **interactive graph manipulation** with node dragging, edge weight editing, and step-by-step algorithm execution. with real-time color highlighting.
- Developed **user authentication system** with **JSON**-based persistent storage for credentials and graph states.

Watch-It https://github.com/Peter-Refaat/Watch-IT — Nov - 2024

- Developed **full-stack Java application** simulating streaming platform functionality with user/admin roles and interactive UI.
- Implemented search system using **dynamic programming** for efficient movie/TV show lookup.
- Built user management system with **regex validation** for credentials and personalized watchlist/rating features.
- Designed responsive **JavaFX interface** with modern UI components and detailed media information displays.

Extracurricular Activities

Noon Cake Programming Contest (NCPC) Feb – 2026

- Achieved 16th place among 800+ participants in the online qualifiers, qualifying for the onsite final round.

DSC Cairo University Programming Contest Mar – 2026

- Advanced through three online rounds among 100+ teams to qualify for the onsite finals, achieving 7th place out of 20+ finalist teams.

HackerRank Problem Solving Basic & Intermediate tests Jul – 2025

- Successfully solved all problems in both assessments (2 per test) within the 90-minute time limit.
- Covered topics including string manipulation, greedy algorithms, and advanced combinatorics using bitmasking and hash maps.

Applai Machine Learning Workshop (Student Activity) Feb – 2025

- Applied core machine learning models including Linear/Logistic Regression, Polynomial Regression, and SVM in practical tasks.
- Ranked 2nd out of 20+ teams in the workshop competition.

acmASCIS Winter Camp Contest (Student Activity) Jan – 2025

- Solved 6 algorithmic problems, achieving 8th place out of 30 participants.

MAM Frontend Summer Training (Student Activity) Sep – 2024

- Gained hands-on experience in frontend development fundamentals using HTML, CSS, and JavaScript.
- Designed and implemented a fully responsive website.